IMU数据打印

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- 1.连接设备
- 2.绑定设备id
  - 2.1设备查看命令
  - 2.2建立端口映射关系
    - 规则文件语法简介
  - 2.3验证查看
- 3.安装驱动库
  - 3.1下载文件
  - 3.2安装驱动库
    - 解压缩
    - 安装库
    - 安装其他必要的库
- 4.IMU的ros环境配置
  - 4.1新建、编译工作空间
  - 4.2打印imu数据

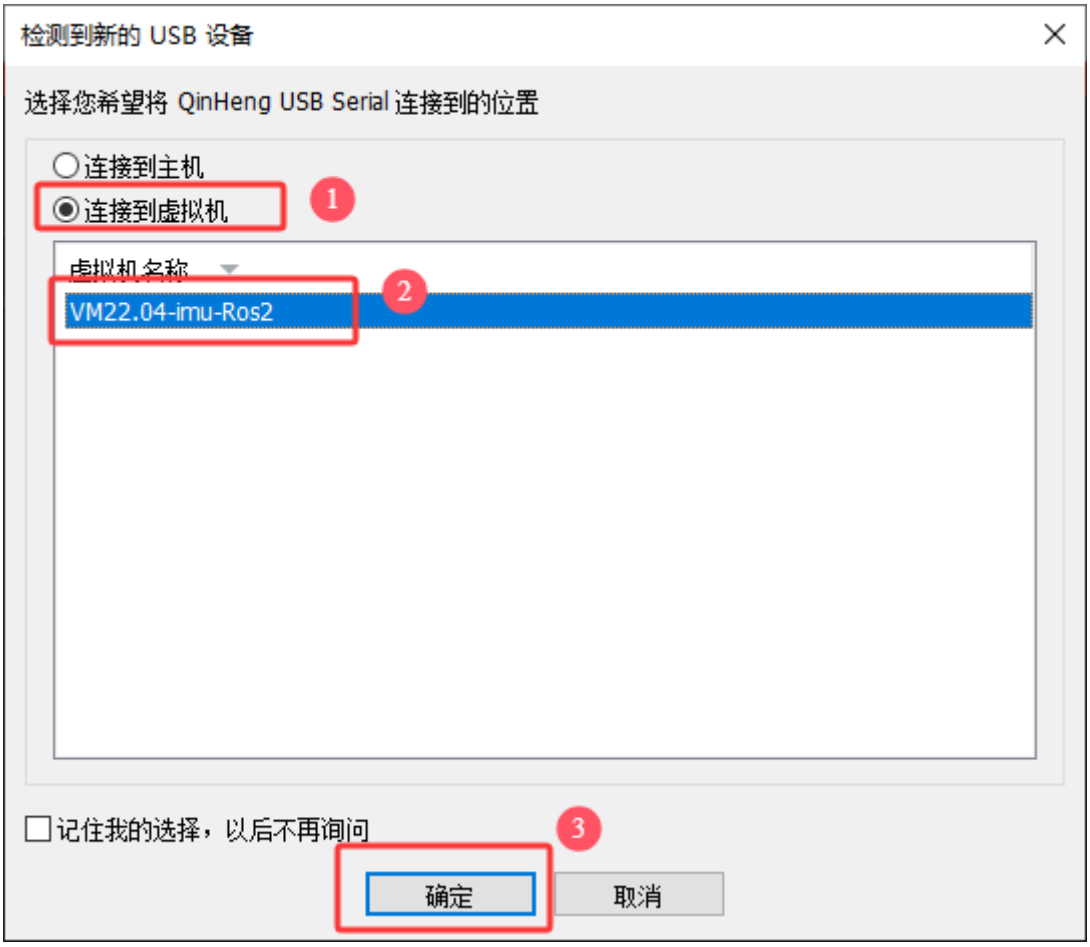
本教程以ubuntu22.04，ros2版本为humble为例。

1.连接设备

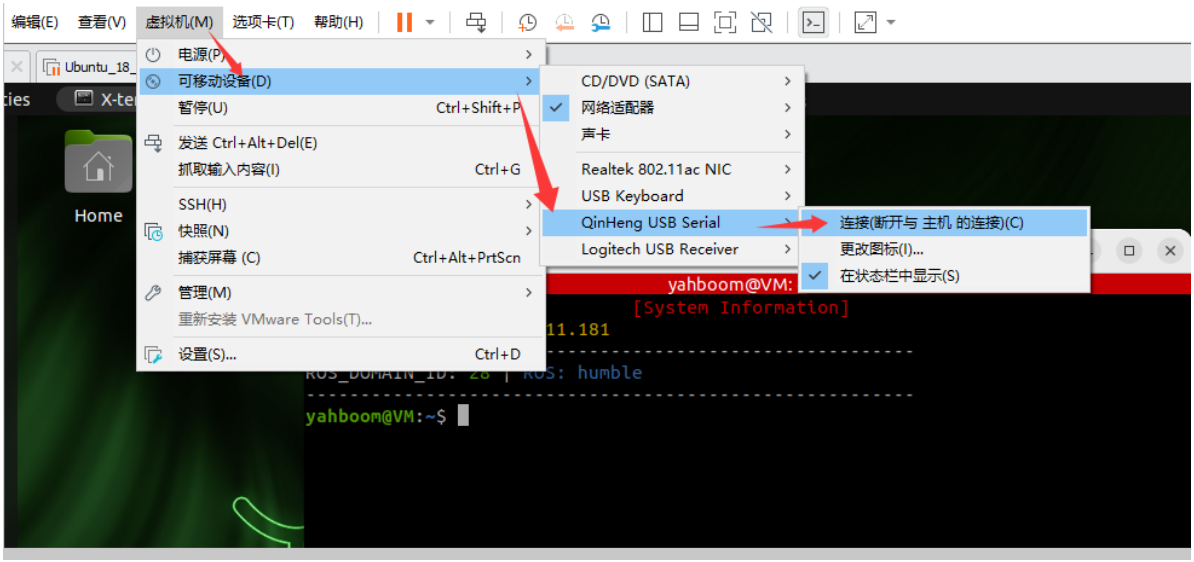
将IMU姿态传感器通过type-c线插在主控的USB上。

如果是使用的虚拟机，参考以下步骤操作

虚拟机有弹窗



虚拟机没有出现弹窗



2.绑定设备id

2.1设备查看命令

设备ID查看

```
lsusb
```

```
yahboom@VM: ~
yahboom@VM: ~ 80x24
                    [System Information]
IP_Address_1: 192.168.11.181
-------------------------------------------------------
ROS_DOMAIN_ID: 28 | ROS: humble
-------------------------------------------------------
yahboom@VM:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 004: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 003: ID 0e0f:0002 VMware, Inc. Virtual USB Hub
Bus 003 Device 005: ID 1a86:7523 QinHeng Electronics CH340 serial converter
Bus 003 Device 002: ID 0e0f:0003 VMware, Inc. Virtual Mouse
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 001 Device 001: ID 1d6b:0001 Linux Foundation 1.1 root hub
yahboom@VM:~$
```

设备号查看

```
ll /dev/ttyU*
```

```
yahboom@VM:~$ ll /dev/ttyU*
crw-rw---- 1 root dialout 188, 0 10月 22 19:00 /dev/ttyUSB0
yahboom@VM:~$
```

2.2建立端口映射关系

进入rules.d目录下

```
cd /etc/udev/rules.d/
```

新建一个myimu.rules文件，并写入以下内容

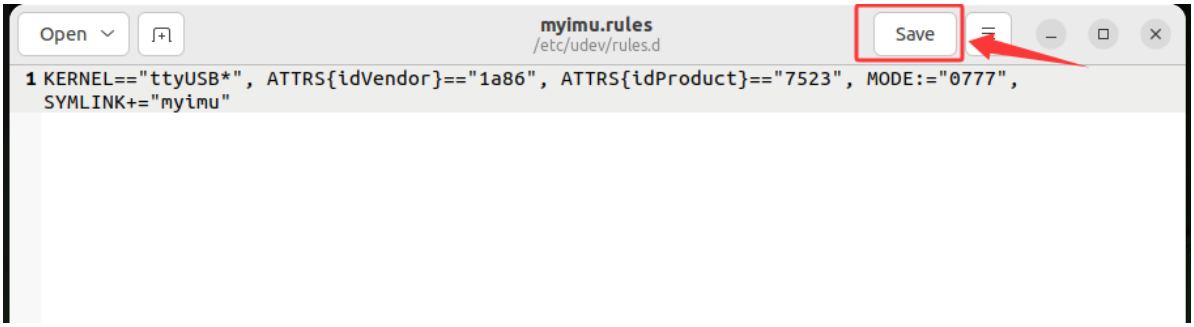
```
sudo gedit myimu.rules
```

如提示找不到gedit命令，运行

```
sudo apt install gedit -y
```

写入内容

```
KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523",
MODE:="0777", SYMLINK+="myimu"
```



保存退出，使规则生效

```
sudo udevadm trigger
sudo service udev reload
sudo service udev restart
```

重新插拔USB设备，大功告成。

规则文件语法简介

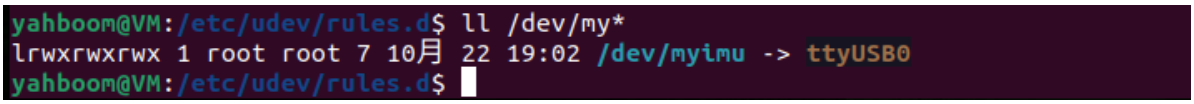
```
KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523",
MODE:="0777", SYMLINK+="myimu"
```

解析

```
KERNEL          # 匹配事件的设备名
ATTR{filename}  # 匹配事件设备的sysfs属性。
idVendor        # 生产商编号
idProduct       # 产品号
SYMLINK         # 为/dev/下的设备文件产生符号链接。就是给这个设备取一个别名。
MODE            # 为设备设定权限。
```

2.3验证查看

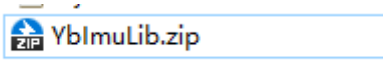
```
ll /dev/my*
```



3.安装驱动库

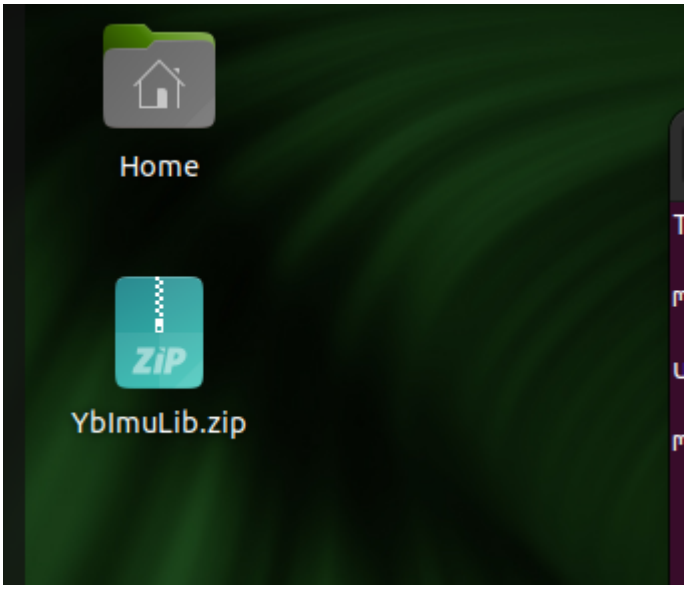
3.1下载文件

资料文件夹有提供最新版本的驱动库，名字为YbImuLib.zip



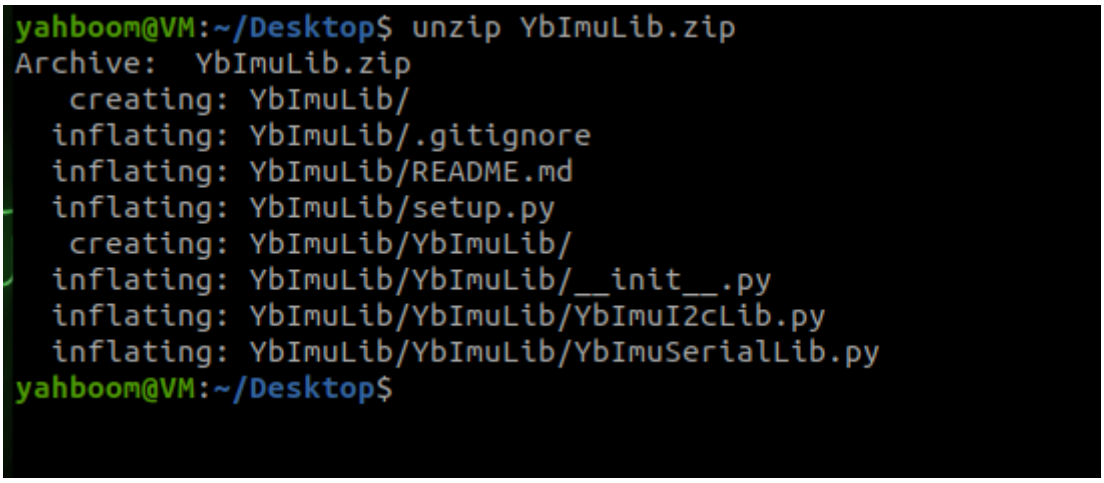
3.2安装驱动库

将YbImuLib.zip文件复制到虚拟机里



解压缩

```
unzip YbImuLib.zip
```

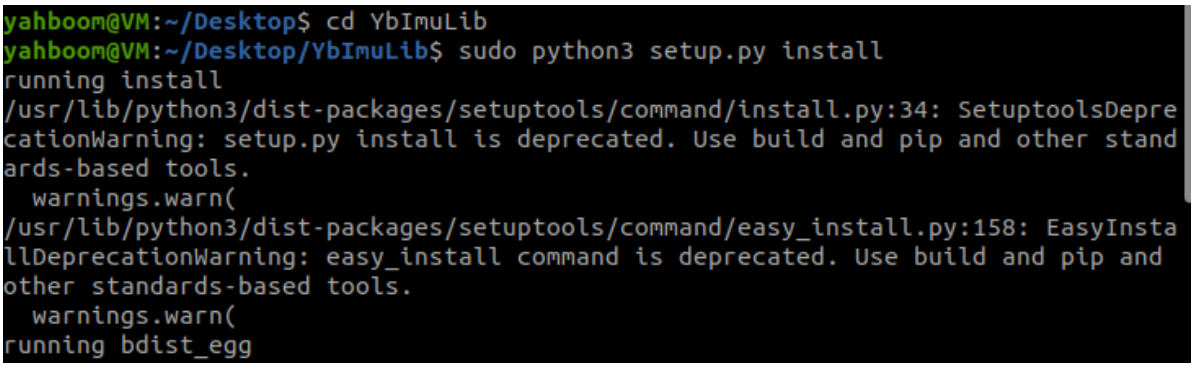


安装库

```
cd YbImuLib
```

运行安装命令，看到最后提示的安装版本号则表示安装成功。此命令会覆盖掉之前已经安装过的Rosmaster_Lib驱动库。

```
sudo python3 setup.py install
```



安装其他必要的库

```
sudo pip3 install pyserial
sudo pip3 install smbus2
```

4.IMU的ros环境配置

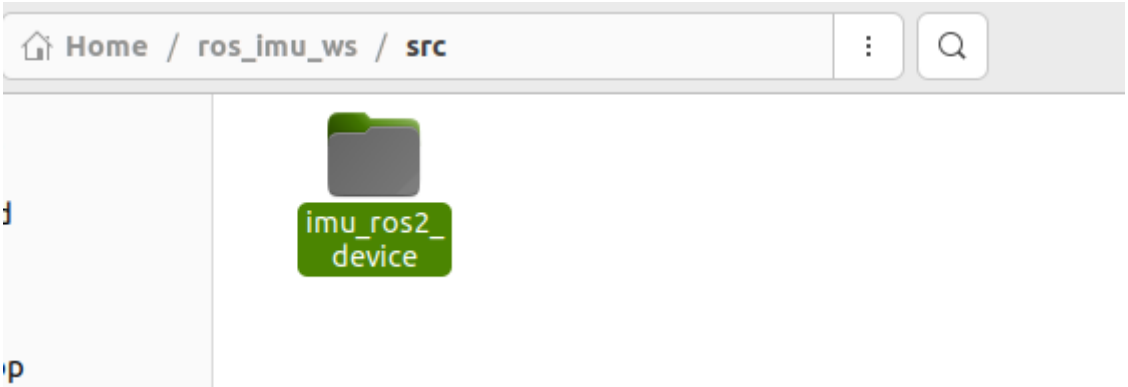
4.1新建、编译工作空间

打开命令终端，运行下面指令：

回到home目录，新建工作空间

```
mkdir ros_imu_ws
cd ros_imu_ws
mkdir src
colcon build --symlink-install
```

将imu_ros2_device文件夹复制到~/ros_imu_ws/src目录下



编译

```
cd ~/ros_imu_ws
colcon build
```

编译通过后，把工作空间的路径添加到.bashrc中，

```
sudo gedit ~/.bashrc
```

把以下内容复制到文件末尾，

```
source ~/ros_imu_ws/install/setup.bash
```

```
# Print IP address 1
echo -e "\033[33mIP_Address_1: $(hostname -I | awk '{print $1}')\033[0m"
# Print IP address 2
#echo -e "\033[33mIP_Address_2: $(hostname -I | awk '{print $2}')\033[0m"

export ROS_DOMAIN_ID=28

# ros2
source /opt/ros/humble/setup.bash

echo "-------------------------------------------------------"
echo -e "ROS_DOMAIN_ID: \033[32m$ROS_DOMAIN_ID\033[0m | \033[34mROS: $(printenv ROS_DISTRO)
\033[0m"
echo "-------------------------------------------------------"

source ~/ros_imu_ws/install/setup.bash
```

保存退出，更新环境变量

```
source  ~/.bashrc
```

4.2打印imu数据

打开一个终端，启动imu节点

```
ros2 run imu_ros2_device ybimu_driver
```

打开第二个终端，检查imu话题

```
ros2 topic list
```

```
yahboom@VM:~$ ros2 topic list
/baro
/euler
/imu/data_raw
/imu/mag
/parameter_events
/rosout
yahboom@VM:~$
```

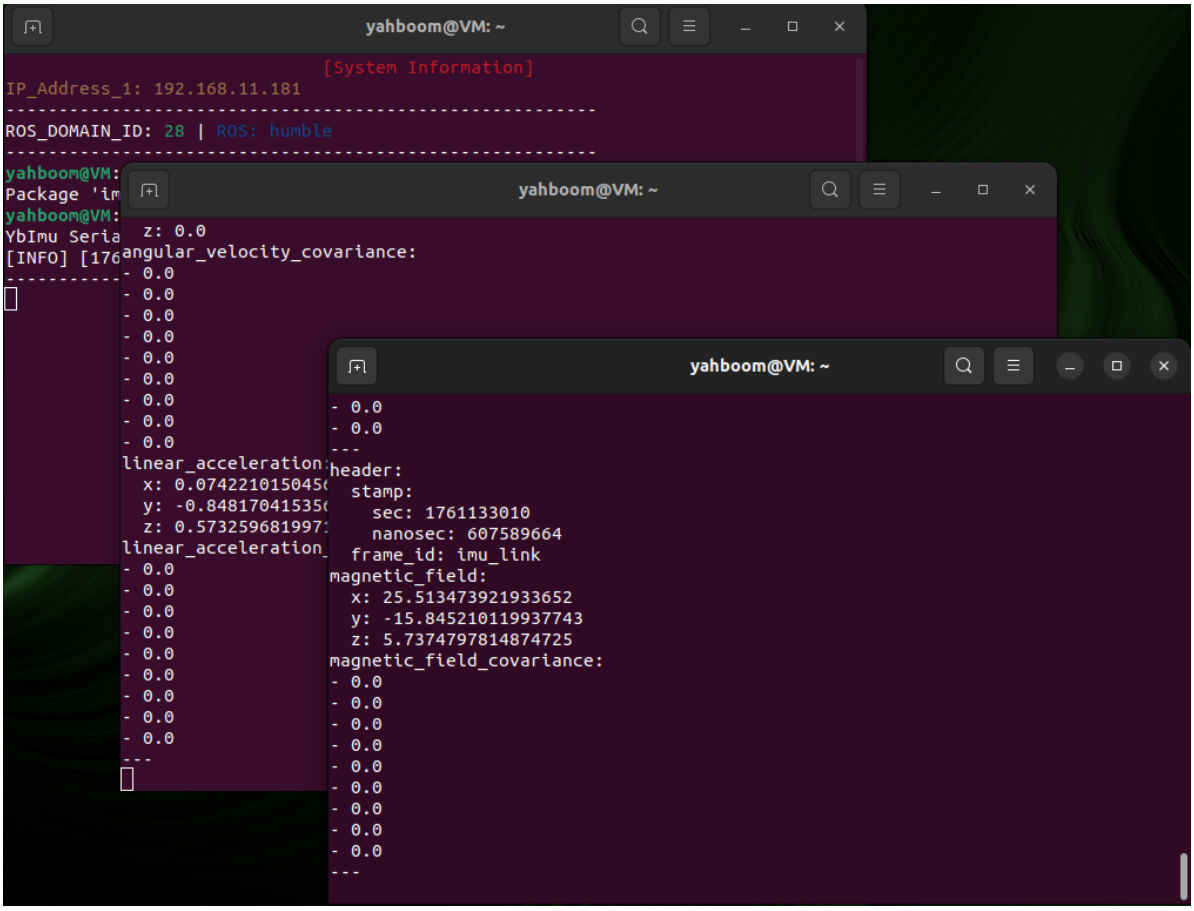
打印/imu/data_raw话题数据

```
ros2 topic echo /imu/data_raw
```

```
header:
  stamp:
    sec: 1761132965
    nanosec: 7432176
  frame_id: imu_link
orientation:
  x: -0.4606093764305115
  y: 0.08095275610685349
  z: -0.22871613502502441
  w: 0.8518229126930237
orientation_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
angular_velocity:
  x: 0.0
  y: 0.0
  z: 0.0
angular_velocity_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
linear_acceleration:
  x: 0.07324442274239326
  y: -0.848170415356914
  z: 0.5727713858455153
linear_acceleration_covariance:
```

新打开第三个终端，打印/imu/mag 话题数据

```
ros2 topic echo /imu/mag
```


yahboom@VM: ~
[System Information]
IP_Address_1: 192.168.11.181
ROS_DOMAIN_ID: 28 | ROS: humble
yahboom@VM: ~
z: 0.0
angular_velocity_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
linear_acceleration:
x: 0.074221015045
y: -0.848170415356
z: 0.573259681997
linear_acceleration_
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
---
yahboom@VM: ~
- 0.0
- 0.0
---
header:
stamp:
sec: 1761133010
nanosec: 607589664
frame_id: imu_link
magnetic_field:
x: 25.513473921933652
y: -15.845210119937743
z: 5.7374797814874725
magnetic_field_covariance:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0